❖ Final RPS (35)

Histology

1- Which of the following sensory receptors in olfactory mucosa:

A) bipolar cells B) multipolar cells

C) squamous cells D) basal cells

2- Apical surface of brush cell has:

A) kinocilia B) stereocilia

C) microvilli D) nothing

3- Dust cells are:

A) alveolar macrophages

Parasitology

4- A patient presents with chest pain and chronic cough. Chest X-ray reveals a lung mass. Chocolate like sauce aspirated from the lesion. What's the most likely diagnosis?

A) Hydatid disease B) Cysticercosis

C) Sparganosis D) Amoebic lung abscess

Anatomy

5- How many arteries in typical intercostal space?

A) one B) two

C) three D) four

6- Type 2 pneumocytes (surfactant secreting cells) appear around:

A) 16 weeks B) 20 weeks

C) 30 weeks D) 24 weeks

7- Which of the following separates oesophagus from trachea:

A) septum transversum B) tracheoesophageal septum

8- Cricothyroid muscle is supplied by:

A) internal branch of superior laryngeal nerve

B) external branch of superior laryngeal nerve

C) recurrent laryngeal nerve

9- Lingula presents in:

A) right upper lobe B) left upper lobe

C) left lower lobe D) right lower lobe

10- Trachea begins at the level of:

A) C6 B) C3

C) C4 D) T1

11- Nasolacrimal duct opens in:

A) spheno-ethmoidal recess B) superior meatus

C) middle meatus D) inferior meatus

12- Rima Glottidis is an opening between:

A) true vocal cords B) false vocal cords C) arytenoid cartilages

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Pathology

13- Alpha 1 anti trypsin deficiency associated with:

A) Panacinar emphysema B) Centriacinar emphysema

C) senile emphysema D) interstitial emphysema

14- Malignant tumor of pleura is:

A) Mesothelioma

B) Bronchogenic carcinoma

15- The lung tissue has a liver-like consistency in:

A) lobar pneumonia B) bronchopneumonia

C) bronchiectasis D) bronchogenic carcinoma

16- Byssinosis is caused by:

A) cotton fibers B) cane sugar fibers

C) coal worker D) silica dust

17- Still birth baby is characterized by:

A) die within days B) premature, diabetic mother

C) hyaline membrane formation D) blue lung with slit like alveolar spaces

18- In viral pneumonia, the inflammatory infiltrate occurs in:

A) alveolar wall B) alveolar lumen C) bronchial wall D) interstitial tissue

19- Bronchial asthma is characterized by:

A) decreased inspiratory volume with normal expiratory volume

B) decreased inspiratory & expiratory volumes

C) normal inspiratory & expiratory volumes

D) decreased expiratory volume with normal inspiratory volume

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Biochemistry

20- All of the following lead to metabolic acidosis Except:

A) Renal failure B) Hypokalemia C) Diarrhea D) DKA

Pharmacology

21- Which of the following anti TB used as prodrug:

A) Isoniazid B) Rifampin C) Ciprofloxacin D) Cycloserine

22- Which of the following is considered expectorant:

A) Ammonium salts

23- One of the following is 5-lipoxygenase enzyme inhibitors used in bronchial asthma:

A) Zileuton B) Zafirlukast C) Montelukast D) Pranlukast

Microbiology

1- One cytokines involved in response to bronchial asthma *(0.5 mark)*

2- Mention 2 clinical conditions caused by aspergillum *(1 mark)*

3- Mention 2 disadvantages for BCG vaccine *(1 mark)*

4- Mention 2 fungal agents that cause systemic mycoses *(1 mark)*

5- Mention 2 types of pneumonia according to **site** of **acquisition** *(1 mark)*

6- In strongly suggestive of diphtheria, specific treatment with ..... should be started immediately *(1 mark)*

7- ..... major changes due to **reassortment of gene** segments that occurs when one cell is infected with **2 different viruses** *(1 mark)*

8- QuantiFERON-TB Gold measures the amount of **....** released from the patient's **WBCs** *(1 mark)*

Physiology

9- V / Q ratio is ..... *(0.5 mark)*

10- Mention 2 safety factors against pulmonary edema ..... *(2 marks)*

11- Mention 2 causes for alveolar recoil .... *(2 marks)*

12- Mention 2 factors that increase P50 in oxygen dissociation curve ..... *(2 marks)*

13- Significance of Haldane effect ..... *(1 mark)*

14- In acclimatization to high altitude, secondary hyperventilation occurs due to ..... *(1 mark)*

15- ..... volume of lung at which the total pulmonary vascular resistance is **typically lowest** *(1 mark)*

16- ..... center is responsible for ramp signals (prevent gasping) *(1 mark)*

Pharmacology

17- Mention 2 side effects for ethambutol (2 marks)

18- Mention 4 risk factors for pneumonia (2 marks)

ANSWERS									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	C	A	D	C	D	B	B	B	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	A	A	A	A	A	D	D	D	B
21	**22**	**23**							
A	A	A							

Microbiology

1. **Tumor necrosis factor (TNF) – Interleukin-3 (IL-3) – Interleukin-4 (IL-4) – Interleukin-5 (IL-5) – Chemokines**
2. **Allergic bronchopulmonary aspergillosis (asthma – type I hypersensitivity) – Aspergilloma (fungus ball) – Pneumonia in immunocompromised patients**
3. **Does not prevent infection – Gives false sense of security – Makes tuberculin test useless in diagnosis especially in latent TB – Not given to immunocompromised children**
4. **Histoplasma capsulatum – Blastomyces dermatitidis – Coccidioides immitis**
5. **Community-acquired pneumonia – Nosocomial pneumonia (hospital-acquired pneumonia) – Ventilator-associated pneumonia**
6. **Diphtheria antitoxic serum**
7. **Antigenic shift**
8. **Interferon-gamma (IFN-γ) released from patient's white blood cells**

Physiology

9. Ratio of amount of air reaching alveoli per minute (V) to amount of blood reaching alveoli per minute (Q)

10. Surfactant that decreases surface tension – Low pulmonary capillary pressure – Negativity of interstitial space – Efficient lymphatic drainage

11. Elastic forces of lung tissue (1/3) – Surface tension of alveolar fluid (2/3)

12. Increase PCO_2 – Increase H^+ concentration (Acidosis) – Increase temperature – Increase 2,3-DPG

13. When HbO2 gives its O2 to the tissues and is reduced to Hb, its capacity to carry CO2 from the tissues is increased. At the lung, oxygenation of Hb to HbO2 decreases its capacity to carry CO2 which is released from venous blood to alveolar air (i.e. increased CO2 removal)

14. increase renal excretion of bicarbonate leading to increased sensitivity of respiratory center to Co2

15. Functional residual capacity

16. Inspiratory center (Dorsal respiratory group)

Pharmacology

17. Optic neuritis - Hyperuricemia - GIT irritation - Peripheral neuropathy

18. Older age – Chronic comorbidities – Immunocompromising conditions – Viral respiratory tract infection – Impaired airway protection (aspiration) – COPD - Smoking and alcohol overuse – Crowded living conditions – Residence in low-income settings